TRIGONOMETRY

त्रिकोणमिति

SHEET-02

CLASS NOTES

If $\alpha + \beta = 90° \Rightarrow \alpha$ and β are complimentary.

$$\tan\alpha \times \tan\beta = 1,\ \cot\alpha \times \cot\beta = 1,$$
$$\sin\alpha \times \sec\beta = 1,\ \cos\alpha \times \operatorname{cosec}\beta = 1$$

$$A + B = 90^\circ$$

$$\Rightarrow A = 90 - B$$

$$\Rightarrow \sin A = \sin(90 - B)$$

$$\Rightarrow \boxed{\sin A = \cos B} \quad \text{— (i)}$$

$$\Rightarrow \sin A \frac{1}{\cos B} = 1$$

$$\Rightarrow \boxed{\sin A \times \sec B = 1} \quad \text{— (ii)}$$

$A + B = 90^\circ$

- $\sin A \sec B = 1$
- $\cos A \operatorname{cosec} B = 1$
- $\tan A \tan B = 1$
- $\cot A \cot B = 1$

Q. $\sin 40^\circ \sec x = 1$

$x = ?$

Ans $40 + x = 90$

$x = 50^\circ$

Q. $\sin 35^\circ \operatorname{cosec} y^\circ = 1$

$y = ?$

Ans $35 + y = 90$

$y = 55^\circ$

Q. $\sin 20 \operatorname{cosec} 70^\circ = ?$

Ans $20 + 70 = 90^\circ$

$\sin 20 \operatorname{cosec} 70 = 1$

1. $2(\sin 1^\circ \times \sec 89^\circ) + 3(\cos 11^\circ \times \operatorname{cosec} 79^\circ) + 5(\tan 21^\circ \times \tan 69^\circ) = ?$

SSC CPO 04/10/2023 (Shift-02)

(a) 20
(b) 12
(c) 11
(d) 10

$2(1) + 3(1) + 5(1)$

$= 10$

2. The value of $\tan5^\circ \tan25^\circ \tan45^\circ \tan65^\circ \tan85^\circ$ is equal to _____.

$\tan5^\circ \tan25^\circ \tan45^\circ \tan65^\circ \tan85^\circ$ का मान _____ के बराबर है।

SSC CGL 24/07/2023 (Shift-2)

(a) 4 (b) 3

(c) 1 (d) 2

$= 1 \times 1 \times 1$

$= 1$

$\tan x \tan y = 1$

$x + y = 90^\circ$

3. The value of cot 13° cot 27° cot 60° cot 63° cot 77° is:

cot 13° cot 27° cot 60° cot 63° cot 77° का मान है:

SSC CHSL 02/08/2023 (Shift-02)

(a) $\frac{1}{\sqrt{3}}$ (b) 0

(c) $\sqrt{3}$ (d) 1

$= 1 \times 1 \times \frac{1}{\sqrt{3}}$

$= \frac{1}{\sqrt{3}}$

M-1

$\frac{1}{\cos\theta} = \sec\theta$

$\frac{1}{\sec\theta} = \cos\theta$

4. **Evaluate:/सरल करें।**

$$7\left(\frac{\operatorname{cosec} 24^\circ}{\sec 66^\circ}\right)^3 + 8\left(\frac{\cot 37^\circ}{\tan 53^\circ}\right)^4 - \left(2\frac{\sec 14^\circ}{\operatorname{cosec} 76^\circ}\right)^2 + \left(-3\frac{\tan 82^\circ}{\cot 8^\circ}\right)^3$$

SSC Phase-XII 20/06/2024 (Shift-02)

(a) 40
(b) 2
(c) −16
(d) 10

$7\left(\cos 66^\circ \operatorname{cosec} 24\right)^3 + 8\left(\cot 53 \cot 37\right)^4 - \left(2\sin 76 \sec 14^\circ\right)^2 + \left(-3 \tan 8 \tan 82\right)^3$

$= 7(1)^3 + 8(1)^4 - (2\times 1)^2 + (-3\times 1)^3$

$= 7 + 8 - 4 - 27$

$= -16$

Complementary

* $\sin\theta \rightleftharpoons \cos\theta$
* $\sec\theta \rightleftharpoons \operatorname{cosec}\theta$
* $\tan\theta \rightleftharpoons \cot\theta$

* $\sin 10 = \cos 80$
* $\sin 20 = \cos 70^\circ$
* $\cos 60^\circ = \sin 30^\circ$
* $\sec 15^\circ = \operatorname{cosec} 75^\circ$

4. Evaluate:/सरल करें।

$$7\left(\frac{\operatorname{cosec} 24^\circ}{\sec 66^\circ}\right)^3 + 8\left(\frac{\cot 37^\circ}{\tan 53^\circ}\right)^4 - \left(2\frac{\sec 14^\circ}{\operatorname{cosec} 76^\circ}\right)^2 + \left(-3\frac{\tan 82^\circ}{\cot 8^\circ}\right)^3$$

SSC Phase-XII 20/06/2024 (Shift-02)

(a) 40
(b) 2
(c) −16
(d) 10

$= 7\times1 + 8\times1 - 4 - 27$
$= -16$

5. Find the value of $\frac{\sec 35^\circ}{\operatorname{cosec} 55^\circ} - \frac{\cos 25^\circ}{\sin 65^\circ}$

$\frac{\sec 35^\circ}{\operatorname{cosec} 55^\circ} - \frac{\cos 25^\circ}{\sin 65^\circ}$ का मान ज्ञात कीजिए।

SSC CGL 11/09/2024 (Shift-01)

(a) 0

(b) 1

(c) $\frac{1}{2}$

(d) 2

$= 1 - 1$

$= 0$

6. What is the value of $\frac{\cos 50^\circ}{\sin 40^\circ} + \frac{3\ \text{cosec}80^\circ}{\sec 10^\circ} - 2$ $\cos 50^\circ . \text{cosec}\ 40^\circ$?

$\frac{\cos 50^\circ}{\sin 40^\circ} + \frac{3\ \text{cosec}80^\circ}{\sec 10^\circ} - 2\ \cos 50^\circ . \text{cosec}\ 40^\circ$ का मान ज्ञात कीजिए।

SSC CGL MAINS (08/08/2022)

(a) 3
(b) 4
(c) 2
(d) 5

$1 + 3 - 2$
$= 2$

7. The value of

$$\frac{\tan 13^\circ \tan 36^\circ \tan 45^\circ \tan 54^\circ \tan 77^\circ}{2\sec^2 60^\circ (\sin^2 60^\circ - 3\cos 60^\circ + 2)}$$ is:

$$\frac{\tan 13^\circ \tan 36^\circ \tan 45^\circ \tan 54^\circ \tan 77^\circ}{2\sec^2 60^\circ (\sin^2 60^\circ - 3\cos 60^\circ + 2)}$$ का मान ज्ञात करें।

SSC CGL 20/08/2021 (Shift- 03)

(a) $-\frac{1}{4}$

(b) $-\frac{1}{10}$

(c) ✓ $\frac{1}{10}$

(d) $\frac{1}{4}$

$$\frac{1\times1\times1}{2\times4\left[\frac{3}{4}-3\times\frac{1}{2}+\frac{2}{1}\right]}$$

$$= \frac{1}{8\left(\frac{3-6+8}{4}\right)}$$

$$= \frac{1}{2\times5} = \frac{1}{10}$$

$\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$

8. $\left[\dfrac{\sec 59^\circ - \sec 59^\circ}{(\sin 18^\circ \div \cos 72^\circ)} \times \dfrac{\tan 26^\circ}{\cot 64^\circ}\right] + \left[\dfrac{\tan 30^\circ + \tan 15^\circ}{(1 - \tan 30^\circ \tan 15^\circ)}\right]$

$=$ _____.

SSC CHSL 05/07/2024 (Shift-04)

(a) 0
(b) –1
(c) 1
(d) 2

$[0] + \tan(30+15)$

$= 0 + \tan 45^\circ$

$= 1$

$A+B=90^\circ$

- $\sin A \cdot \sec B = 1$
- $\cos A \cdot \operatorname{cosec} B = 1$
- $\tan A \tan B = 1$
- $\cot A \cdot \cot B = 1$

$\sin A = \frac{1}{\sec B}$

$\operatorname{cosec} B = \frac{1}{\cos A}$

$\tan A = \frac{1}{\tan B}$

$\Rightarrow$

$\sin A = \cos B$

$\sec A = \operatorname{cosec} B$

$\tan A = \cot B$

Ex: $\sin x = \cos(x+30)$

$x = ?$

Ans

$x + x + 30 = 90$

$2x = 60$ 30

$x = 30^\circ$

Ex:- $\tan(2x+10) = \cot(3x+10)$

$x = ?$

$2x + 10 + 3x + 10 = 90^\circ$

$5x = 70$ 14

$x = 14^\circ$

9. If $\sin(5x - 40°) = \cos(5y + 40°)$, then the value of $x + y$ is:

यदि $\sin(5x - 40°) = \cos(5y + 40°)$ तो $x + y$ का मान है:

SSC Phase-XII 24/06/2024 (Shift-03)

(a) 18°
(b) 40°
(c) 15°
(d) 20°

$5x - 40 + 5y + 40 = 90$

$5(x + y) = 90$

18

10. If $\sin(x + y) = \cos\{3(x + y)\}$, then, the value of $\cot\{2(x + y)\}$ is _____.

यदि $\sin(x + y) = \cos\{3(x + y)\}$ तो $\cot\{2(x + y)\}$ का मान है।

SSC CPO 29/06/2024 (Shift-02)

(a) –1 (b) 1

(c) –2 (d) 2

$x+y+3x+3y = 90^\circ$

$\Rightarrow 2(x+y) = 90^\circ$

$\Rightarrow 2(x+y) = 45^\circ$

$\cot 45^\circ = 1$

11. If $\tan 2\theta = \cot(\theta - 36)^\circ$, where 2θ is an acute angle, then the value of θ is:

यदि $\tan 2\theta = \cot(\theta - 36)^\circ$, जहाँ 2θ एक न्यून कोण है, तो θ का मान ज्ञात कीजिए।

SSC CHSL 11/08/2023 (Shift-2)

(a) 18°
(b) 36°
(c) 30°
(d) 42°

$2\theta + \theta - 36 = 90$

$\Rightarrow 3\theta = 126$

$\theta = 42^\circ$

12. If $\cos(2\theta + 54)^\circ = \sin\theta$, $0^\circ < (2\theta + 54)^\circ < 90^\circ$, then what is the value of $\dfrac{1}{\cot 5\theta + \sec\dfrac{5\theta}{2}}$?

यदि $\cos(2\theta + 54)^\circ = \sin\theta$, $0^\circ < (2\theta + 54)^\circ < 90^\circ$ है तो $\dfrac{1}{\cot 5\theta + \sec\dfrac{5\theta}{2}}$ का मान ज्ञात कीजिए।

SSC CGL 20/08/2021 (Shift- 02)

(a) $\dfrac{\sqrt{3}}{2}$

(b) $\dfrac{1}{3}$

(c) $\dfrac{\sqrt{3}}{3}$

(d) $2\sqrt{3}$

$3\theta + 54 = 90^\circ$

$3\theta = 36^\circ$ 12.

$\theta = 12^\circ$

$\dfrac{1}{\cot 60^\circ + \sec 30^\circ}$

$= \dfrac{1}{\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}}} = \dfrac{1}{\frac{3}{\sqrt{3}}}$

$= \dfrac{\sqrt{3}}{3}$

13. If $\sin(5x-25)^\circ = \cos(5y+25)^\circ$, where $5x-25^\circ$ and $5y+25^\circ$ are acute angles, then the value of $(x+y)$ is:

यदि $\sin(5x-25)^\circ = \cos(5y+25)^\circ$ है, जहाँ $5x-25^\circ$ और $5y+25^\circ$ न्यूनकोण है, तो $(x+y)$ का मान क्या होगा।

SSC CGL 18/07/2023 (Shift-03)

(a) 50°
(b) 40°
(c) 18°
(d) 16°

$5x-25+5y+25=90$

$5(x+y)=90$

$x+y=18$.

14. If $\tan(5\theta - 10)^\circ = \cot(5\varphi + 20)^\circ$, then the value of $\theta + \varphi$ is :

यदि $\tan(5\theta - 10)^\circ = \cot(5\varphi + 20)^\circ$ है, तो $\theta + \varphi$ का मान क्या है?

SSC CHSL 10/08/2023 (Shift-2)

(a) 16°
(b) 18°
(c) 15°
(d) 20°

$5\theta - 10 + 5\phi + 20 = 90^\circ$

$\Rightarrow 5(\theta + \phi) = 80^\circ$ 16°

15. If $\sec(5\alpha-15)^\circ = \operatorname{cosec}(15-2\alpha)^\circ$, then the value of $\cos\alpha + \sin 2\alpha + \tan(1.5\alpha)$ is:

यदि $\sec(5\alpha-15)^\circ = \operatorname{cosec}(15-2\alpha)^\circ$ है तो $\cos\alpha + \sin 2\alpha + \tan(1.5\alpha)$ का मान ज्ञात कीजिए।

SSC CGL 20/08/2021 (Shift- 01)

(a) $\sqrt{2}+1$

(b) $\sqrt{2}-1$

(c) $\sqrt{3}-1$

(d) $\sqrt{3}+1$

$5\alpha - 15 + 15 - 2\alpha = 90^\circ$

$3\alpha = 90^\circ$

$\alpha = 30^\circ$

$\cos 30 + \sin 60 + \tan 45$

$= \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + 1$

$= \sqrt{3} + 1$

$20+x+60=90^\circ$

$x=10^\circ$

16. If $\sin(20 + x)^\circ = \cos 60^\circ$, $0^\circ \leq (20 + x)^\circ \leq 90^\circ$, then find the value of $2\sin^2(3x + 15)^\circ - \text{cosec}^2(2x + 10)^\circ$.

यदि $\sin(20 + x)^\circ = \cos 60^\circ$, $0^\circ \leq (20 + x)^\circ \leq 90^\circ$ है तो $2\sin^2(3x + 15)^\circ - \text{cosec}^2(2x + 10)^\circ$ का मान ज्ञात कीजिए।

SSC CGL 20/08/2021 (Shift- 01)

(a) 3

(b) −3

(c) −2

(d) $-\frac{1}{3}$

$2\sin^2 45 - \text{cosec}^2 30^\circ$

$= 2 \times \frac{1}{2} - 4$

$= -3$

$3A+A+10=90^\circ$

$4A=80^\circ$ ~~20°~~

ans $= 2\operatorname{cosec}30^\circ+6\tan^2 60^\circ-\frac{3}{2}\tan^2 60^\circ$

$= 2\times2+6\times3-\frac{3}{2}\times3$

$= 4+18-\frac{9}{2}$

$= 22-\frac{9}{2} = \frac{35}{2}$

17. If $\sin3A = \cos(A+10)^\circ$, where 3A is an acute angle then what is the value of

$$2\operatorname{cosec}\frac{3A}{2}+6\tan^2 3A-\frac{3}{2}\tan^2 3A?$$

यदि $\sin3A = \cos(A+10)^\circ$ जहां 3A न्यूनकोण है तो

$$2\operatorname{cosec}\frac{3A}{2}+6\tan^2 3A-\frac{3}{2}\tan^2 3A$$ का मान ज्ञात कीजिए।

SSC CGL 2019 Tier-II (15/10/2020)

(a) $\frac{7}{4}$

(b) $\frac{35}{2}$

(c) $\frac{17}{2}$

(d) 5

18. If $\cos\theta = \sin(2\theta) \neq 0$, what is the value of $\cos^4\theta + \sin^4\theta + \cos^3\theta + \sin^3\theta + \sin^2\theta + \cos^2\theta + \sin\theta + \cos\theta$?

यदि $\cos\theta = \sin(2\theta) \neq 0$, तो $\cos^4\theta + \sin^4\theta + \cos^3\theta + \sin^3\theta + \sin^2\theta + \cos^2\theta + \sin\theta + \cos\theta$ का मान क्या?

SSC Phase X 05/08/2022 (Shift- 03)

(a) $\frac{18+8\sqrt{3}}{7}$

(b) $\frac{8+7\sqrt{3}}{18}$

(c) $\frac{7+18\sqrt{3}}{8}$

(d) $\frac{18+7\sqrt{3}}{8}$

$3\theta = 90^\circ \quad 30^\circ$

$$\left(\frac{\sqrt{3}}{2}\right)^4 + \left(\frac{1}{2}\right)^4 + \left(\frac{\sqrt{3}}{2}\right)^3 + \left(\frac{1}{2}\right)^3 + \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \frac{\sqrt{3}}{2} + \frac{1}{2}$$

$$= \frac{9}{16} + \frac{1}{16} + \frac{3\sqrt{3}}{8} + \frac{1}{8} + \frac{3}{4} + \frac{1}{4} + \frac{\sqrt{3}}{2} + \frac{1}{2}$$

$$= \frac{9+1+6\sqrt{3}+2+12+4+8\sqrt{3}+8}{16}$$

$$= \frac{36+14\sqrt{3}}{16} = \frac{18+7\sqrt{3}}{8}$$

$\sin 30^\circ = \frac{1}{2}$

$\cos 30^\circ = \frac{\sqrt{3}}{2}$

If $\alpha + \beta = 90°$, then

$\sin^2\alpha + \sin^2\beta = 1$ &

$\cos^2\alpha + \cos^2\beta = 1$

$$A+B=90^\circ$$

* $\sin^2A+\cos^2A=1$

$\Rightarrow \sin^2A+\cos^2(90-B)=1$

$\Rightarrow \boxed{\sin^2A+\sin^2B=1}$

and

$$\boxed{\cos^2A+\cos^2B=1}$$

Ex:- $\sin^2 30^\circ+\sin^2 60^\circ$

$= \frac{1}{4}+\frac{3}{4}$

$= \frac{4}{4} = 1$

* $\sin^2 10^\circ+\sin^2 80^\circ$

$= 1$

19. Simplify the given expression.

दिए गए व्यंजक को सरल कीजिए।

$$\left\{\frac{\sin^2 32^\circ + \sin^2 58^\circ}{\cos^2 32^\circ + \cos^2 58^\circ} + \sin^2 53^\circ + \cos 53^\circ \sin 37^\circ\right.$$

SSC CHSL 07/08/2023 Shift-04

(a) 2 (b) −1

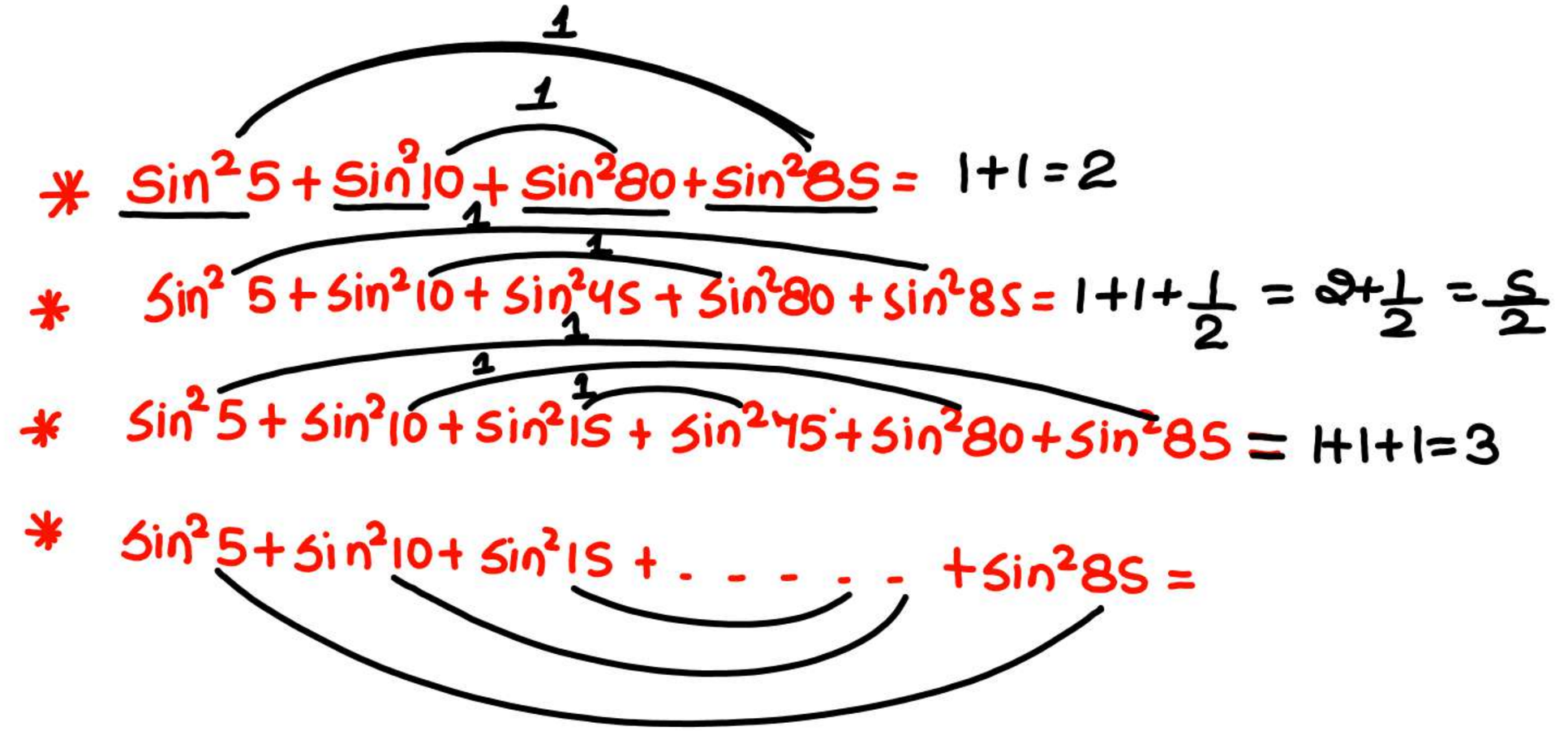
(c) −2 (d) 1

$$= \frac{1}{1} + \sin^2 53^\circ + \sin 37^\circ \cdot \sin 37^\circ$$

$$= 1 + \sin^2 53^\circ + \sin^2 37^\circ$$

$$= 1 + 1$$

$$= 2$$


* $\sin^2 5 + \sin^2 10 + \sin^2 80 + \sin^2 85 = 1+1 = 2$
* $\sin^2 5 + \sin^2 10 + \sin^2 45 + \sin^2 80 + \sin^2 85 = 1+1+\frac{1}{2} = 2+\frac{1}{2} = \frac{5}{2}$
* $\sin^2 5 + \sin^2 10 + \sin^2 15 + \sin^2 45 + \sin^2 80 + \sin^2 85 = 1+1+1 = 3$
* $\sin^2 5 + \sin^2 10 + \sin^2 15 + \ldots\ldots + \sin^2 85 =$

* $\sin^2 5 + \sin^2 10 + \sin^2 80 + \sin^2 85 = \frac{4}{2} = 2$

* $\sin^2 5 + \sin^2 10 + \sin^2 45 + \sin^2 80 + \sin^2 85 = \frac{5}{2}$

* $\sin^2 5 + \sin^2 10 + \sin^2 15 + \sin^2 45 + \sin^2 80 + \sin^2 85 = \frac{6}{2} = 3$

* $\sin^2 5 + \sin^2 10 + \sin^2 15 + \ldots\ldots + \sin^2 85 = \frac{17}{2}$

$$n = \frac{85-5}{5} + 1 = 17$$

Exception

* $\sin^2 5 + \sin^2 10 + \sin^2 15 + \ldots\ldots + \sin^2 85 + \underbrace{\sin^2 90}_{1} = 17 \times \frac{1}{2} + 1 = \frac{19}{2}$

* $\underbrace{\sin^2 0}_{0} + \sin^2 5 + \sin^2 10 + \sin^2 15 + \ldots\ldots + \sin^2 85 = 0 + \frac{17}{2} = \frac{17}{2}$

Exception

* $\underbrace{\sin^2 5 + \sin^2 10 + \sin^2 15 + \cdots + \sin^2 85}_{17} + \underbrace{\sin^2 90}_{1} = 17 \times \frac{1}{2} + 1 = \frac{19}{2}$

* $\underbrace{\sin^2 0}_{0} + \underbrace{\sin^2 5 + \sin^2 10 + \sin^2 15 + \cdots + \sin^2 85}_{17} = 0 + \frac{17}{2} = \frac{17}{2}$

* $\underbrace{\sin^2 0}_{0} + \underbrace{\sin^2 5 + \sin^2 10 + \cdots + \sin^2 80 + \sin^2 85}_{\frac{17}{2}} + \underbrace{\sin^2 90}_{1} =$

$0 + \frac{17}{2} + 1 = \frac{19}{2}$

$\frac{19}{2}$

20. The value of $\sin^2 15° + \sin^2 25° + \sin^2 35° + \sin^2 45° + \sin^2 55° + \sin^2 65° + \sin^2 75°$ is:

$\sin^2 15° + \sin^2 25° + \sin^2 35° + \sin^2 45° + \sin^2 55° + \sin^2 65° + \sin^2 75°$ का मान है।

SSC CPO 29/06/2024 (Shift-03)

(a) 4 (b) 7

(c) $\frac{7}{2}$ (d) $\frac{7}{3}$

$\sin^2 15° + \sin^2 25° + \sin^2 35° + \sin^2 45° + \sin^2 55° + \sin^2 65° + \sin^2 75°$

ans = $\frac{7}{2}$

Q. $\cos^2 10 + \cos^2 20 + \cos^2 45 + \cos^2 70 + \cos^2 80 = \frac{5}{2}$

21. The value of $\frac{5\cos^2 62^\circ + 5\cos^2 28^\circ - 21}{7\sin^2 35^\circ + 7\sin^2 55^\circ + 1}$ is:

$\frac{5\cos^2 62^\circ + 5\cos^2 28^\circ - 21}{7\sin^2 35^\circ + 7\sin^2 55^\circ + 1}$ का मान ज्ञात कीजिए।

SSC CGL 20/04/2022 (Shift- 03)

(a) 3
(b) – 2
(c) 2
(d) – 3

$$= \frac{5(\cos^2 62 + \cos^2 28) - 21}{7(\sin^2 35 + \sin^2 55) + 1}$$

$$= \frac{5 \times 1 - 21}{7 \times 1 + 1} = \frac{-16}{8} = -2$$

22. The value of $\frac{\sqrt{2}\tan(60-\theta)^\circ \tan(30+\theta)^\circ}{\sin^2(45+\theta)^\circ + \sin^2(45-\theta)^\circ}$:

$\frac{\sqrt{2}\tan(60-\theta)^\circ \tan(30+\theta)^\circ}{\sin^2(45+\theta)^\circ + \sin^2(45-\theta)^\circ}$ का मान ज्ञात कीजिए।

SSC CGL 17/08/2021 (Shift- 02)

(a) $\frac{1}{\sqrt{2}}$

(b) 1

(c) 2

(d) $\sqrt{2}$

$\frac{\sqrt{2} \times 1}{1}$

23. The value of $(\sin 37^\circ \cos 53^\circ + \cos 37^\circ \sin 53^\circ) - \dfrac{4\cos^2 37^\circ - 7 + 4\cos^2 53^\circ}{\tan^2 47^\circ + 4 - \operatorname{cosec}^2 43^\circ}$ is:

$(\sin 37^\circ \cos 53^\circ + \cos 37^\circ \sin 53^\circ) - \dfrac{4\cos^2 37^\circ - 7 + 4\cos^2 53^\circ}{\tan^2 47^\circ + 4 - \operatorname{cosec}^2 43^\circ}$ का मान ज्ञात कीजिए।

SSC CGL 18/08/2021 (Shift- 01)

(a) 1 (b) −2

(c) 0 (d) 2

$(\sin^2 37 + \sin^2 53) - \left(\dfrac{4-7}{\tan^2 47 + 4 - \sec^2 47}\right)$

$= 1 - \dfrac{-3}{-1+4}$

$= 1 - \left(\dfrac{-3}{3}\right)$

$= 1 + 1$

$= 2$

$\sec^2\theta - \tan^2\theta = 1$

or

$\tan^2\theta - \sec^2\theta = -1$

23. The value of $(\sin 37^\circ \cos 53^\circ + \cos 37^\circ \sin 53^\circ) - \dfrac{4\cos^2 37^\circ - 7 + 4\cos^2 53^\circ}{\tan^2 47^\circ + 4 - \operatorname{cosec}^2 43^\circ}$ is:

$(\sin 37^\circ \cos 53^\circ + \cos 37^\circ \sin 53^\circ) - \dfrac{4\cos^2 37^\circ - 7 + 4\cos^2 53^\circ}{\tan^2 47^\circ + 4 - \operatorname{cosec}^2 43^\circ}$ का मान ज्ञात कीजिए।

SSC CGL 18/08/2021 (Shift- 01)

(a) 1
(b) −2
(c) 0
(d) 2

$\sin 90 - \dfrac{4-7}{-1+4}$

$= 1 - \left(\dfrac{-3}{3}\right)$

$= 2$

23. The value of $(\sin 37^\circ \cos 53^\circ + \cos 37^\circ \sin 53^\circ) - \dfrac{4\cos^2 37^\circ - 7 + 4\cos^2 53^\circ}{\tan^2 47^\circ + 4 - \operatorname{cosec}^2 43^\circ}$ is:

$(\sin 37^\circ \cos 53^\circ + \cos 37^\circ \sin 53^\circ) - \dfrac{4\cos^2 37^\circ - 7 + 4\cos^2 53^\circ}{\tan^2 47^\circ + 4 - \operatorname{cosec}^2 43^\circ}$ का मान ज्ञात कीजिए।

$\sec^2 47$

SSC CGL 18/08/2021 (Shift- 01)

(a) 1 (b) −2

(c) 0 (d) 2

$= \sin(37+53) - \left\{\dfrac{4(\cos^2 37 + \cos^2 53) - 7}{\tan^2 47 - \sec^2 47 + 4}\right\}$

$= \sin 90 - \left\{\dfrac{4-7}{-1+4}\right\}$

$= 1 - \left(\dfrac{-3}{3}\right)$

$= 1 - (-1)$

$= 1 + 1$

$= 2$

$\sin(x+y) = \sin x \cos y + \cos x \sin y$

* $\sin^2\theta + \cos^2\theta = 1$

* $\sec^2\theta - \tan^2\theta = 1$

$\tan^2\theta - \sec^2\theta = -1$

* $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

$\cot^2\theta - \operatorname{cosec}^2\theta = -1$

24. The value of

$$\frac{4\tan^2 30^\circ + \sin^2 30^\circ \cos^2 45^\circ + \sec^2 48^\circ - \cot^2 42^\circ}{\cos 37^\circ \sin 53^\circ + \sin 37^\circ \cos 53^\circ + \tan 18^\circ \tan 72^\circ}$$

का मान ज्ञात कीजिए।

SSC CGL MAINS 29/01/2022

(a) $\frac{35}{48}$

(b) $\frac{59}{48}$

(c) $\frac{49}{24}$

(d) $\frac{35}{24}$

$$\frac{4\times\frac{1}{3}+\frac{1}{4}\times\frac{1}{2}+\frac{1}{1}}{1+1}=\frac{\frac{32+3+24}{24}}{2}$$

$$=\frac{59}{48}$$

$\sec^2 48^\circ - \tan^2 48^\circ$

$\sin(A+B) = \sin A\cos B + \cos A \sin B$

25. The value of $\frac{5\cos^2 62^\circ + 5\cos^2 28^\circ - 21}{7\sin^2 35^\circ + 7\sin^2 55^\circ + 1}$ is:

$\frac{5\cos^2 62^\circ + 5\cos^2 28^\circ - 21}{7\sin^2 35^\circ + 7\sin^2 55^\circ + 1}$ का मान ज्ञात कीजिए।

SSC CGL 20/04/2022 (Shift- 03)

(a) 3
(b) – 2
(c) 2
(d) – 3

$\frac{5\times 1 - 21}{7+1} = -\frac{16}{8} = -2$

26. The value of

$$\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\operatorname{cosec}^2 71^\circ + \cos^2 15^\circ - \tan^2 19^\circ + \cos^2 75^\circ}$$ **is:**

$$\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\operatorname{cosec}^2 71^\circ + \cos^2 15^\circ - \tan^2 19^\circ + \cos^2 75^\circ}$$ का मान ज्ञात कीजिए।

SSC CGL MAINS 03/02/2022

(a) 1 (b) 4

(c) -3 (d) 2

$$\frac{2\cos 9^\circ \times 2\sec 9^\circ}{1+1} = \frac{4}{2} = 2$$

$1+\tan^2\theta=\sec^2\theta$

27. If $x = \sec 57°$, then $\cot^2 33° + \sin^2 57° + \sin^2 33° + \text{cosec}^2 57° \cos^2 33° + \sec^2 33° . \sin^2 57°$ is equal to :

यदि $x = \sec 57°$ है तो $\cot^2 33° + \sin^2 57° + \sin^2 33° + \text{cosec}^2 57° \cos^2 33° + \sec^2 33° . \sin^2 57°$ का मान ज्ञात कीजिए।

SSC CGL TIER-II (16/10/2020)

(a) $x^2 + 2$

(b) $2x^2 + 1$

(c) $x^2 + 1$

(d) $\dfrac{1}{x^2+1}$

$\tan^2 57 + 1 + 1 + 1$

$= \sec^2 57 + 2$

$= x^2 + 2$

28. Find the value of the following.

निम्नलिखित का मान ज्ञात कीजिए।

$$\frac{\sin 67^\circ \cos 37^\circ - \sin 37^\circ \cos 67^\circ}{\cos 13^\circ \cos 17^\circ - \sin 13^\circ \sin 17^\circ}$$

SSC CHSL 27/05/2022 (Shift-03)

(a) $\frac{1}{\sqrt{3}}$

(b) $\frac{4}{\sqrt{3}}$

(c) $\frac{2}{\sqrt{3}}$

(d) 7

$$\frac{\sin(67-37)}{\cos(13+17)} = \frac{\sin 30}{\cos 30} = \tan 30 = \frac{1}{\sqrt{3}}$$

29. The value of $\frac{\sin^2 52^\circ + 2 + \sin^2 38^\circ}{4\cos^2 43^\circ - 5 + 4\cos^2 47^\circ}$ is:

$\frac{\sin^2 52^\circ + 2 + \sin^2 38^\circ}{4\cos^2 43^\circ - 5 + 4\cos^2 47^\circ}$ का मान ज्ञात करें।

$4(\cos^2 43 + \cos^2 47) - 5$

SSC CPO 24/11/2020 (Shift-2)

(a) 3

(b) $\frac{1}{3}$

(c) $-\frac{1}{3}$

(d) −3

$= \frac{1+2}{4-5} = \frac{3}{-1} = -3$

$$\frac{2\times 1+\sin^2 26+\sin^2 64}{-1-1}$$

$$=\frac{2+1}{-2}=-\frac{3}{2}$$

30. The value of

$$\frac{2\sin^2 38^\circ \sec^2 52^\circ+\cos 64^\circ \sin 26^\circ+\sin^2 64^\circ}{\tan^2 23^\circ+\cot^2 23^\circ-\sec^2 67^\circ-\operatorname{cosec}^2 67^\circ}$$

is :

$$\frac{2\sin^2 38^\circ \sec^2 52^\circ+\cos 64^\circ \sin 26^\circ+\sin^2 64^\circ}{\tan^2 23^\circ+\cot^2 23^\circ-\sec^2 67^\circ-\operatorname{cosec}^2 67^\circ}$$

का मान ज्ञात कीजिए।

SSC CGL 2019 TIER-II (15/10/2020)

(a) -2

(b) $\frac{3}{2}$

(c) 2

(d) $-\frac{3}{2}$

$$\frac{3\times1-1}{\tan^2(28-\phi)-\sec^2(28-\phi)}$$

$$= \frac{2}{-1} = -2$$

31. The value of

$$\frac{3(\underbrace{\cos ec^2 26^\circ}_{\sec^2 64} - \tan^2 64^\circ) + (\cot^2 42^\circ - \underbrace{\sec^2 48^\circ}_{\operatorname{cosec}^2 42})}{\cot(22-\varnothing)^\circ - \cos ec^2(62+\varnothing)^\circ - \tan(\varnothing+68)^\circ + \tan^2(28-\varnothing)^\circ}$$

$$+ \frac{\tan(68+\phi) - \sec^2(28-\phi)}{3(\cos ec^2 26^\circ - \tan^2 64^\circ) + (\cot^2 42^\circ - \sec^2 48^\circ)}{\cot(22-\varnothing)^\circ - \cos ec^2(62+\varnothing)^\circ - \tan(\varnothing+68)^\circ + \tan^2(28-\varnothing)^\circ}$$

का मान ज्ञात कीजिए।

SSC CGL MAINS 03/02/2022

(a) 3 (b) 4

(c) −1 (d) −2

$\sec^2\theta - \tan^2\theta = 1$ or $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

$\tan^2\theta - \sec^2\theta = -1$ or $\cot^2\theta - \operatorname{cosec}^2\theta = -1$

$$\frac{3(-1)-2(-1)}{\sec^2(22-\theta)-\tan^2(22-\theta)}$$

$$= \frac{-3+2}{1} = \frac{-1}{1} = -1$$

32. The value of

$$\frac{3(\cot^2 47^\circ - \sec^2 43^\circ) - 2(\tan^2 23^\circ - \operatorname{cosec}^2 67^\circ)}{\operatorname{cosec}^2(68+\theta)^\circ - \tan(\theta+61)^\circ - \tan^2(22-\theta)^\circ + \cot(29-\theta)^\circ}$$

$$\frac{3(\cot^2 47^\circ - \sec^2 43^\circ) - 2(\tan^2 23^\circ - \operatorname{cosec}^2 67^\circ)}{\operatorname{cosec}^2(68+\theta)^\circ - \tan(\theta+61)^\circ - \tan^2(22-\theta)^\circ + \cot(29-\theta)^\circ}$$

का मान ज्ञात कीजिए।

SSC CGL MAINS 29/01/2022

(a) 1 (b) 0

(c) 5 (d) −1

33. The value of/का मान बताइए।

$$\frac{\sin23^\circ\cos67^\circ + \sec52^\circ\sin38^\circ + \cos23^\circ\sin67^\circ + \operatorname{cosec}52^\circ\cos38^\circ}{\operatorname{cosec}^2 20^\circ - \tan^2 70^\circ}$$

SSC CGL 11/04/2022 (Shift- 01)

(a) 3 (b) 4

(c) 2 (d) 0

$$\frac{\sin^2 23 + 1 + \sin^2 67 + 1}{1}$$

$$= \frac{1+2}{1} = 3$$